

CircuitFLow

Hardware and Budget Analysis

PHASE - 2

Table of Contents

Table of Contents 2

Applicable Hardware Components 3

Estimated Budget 3

Hardware Connections 4

Applicable Hardware Components

a. Arduino UNO MCU

b. Power supply and data extraction via USB-B

c. YF-S401 water-flow sensor

d. Standard jumper wires

e. Resistors

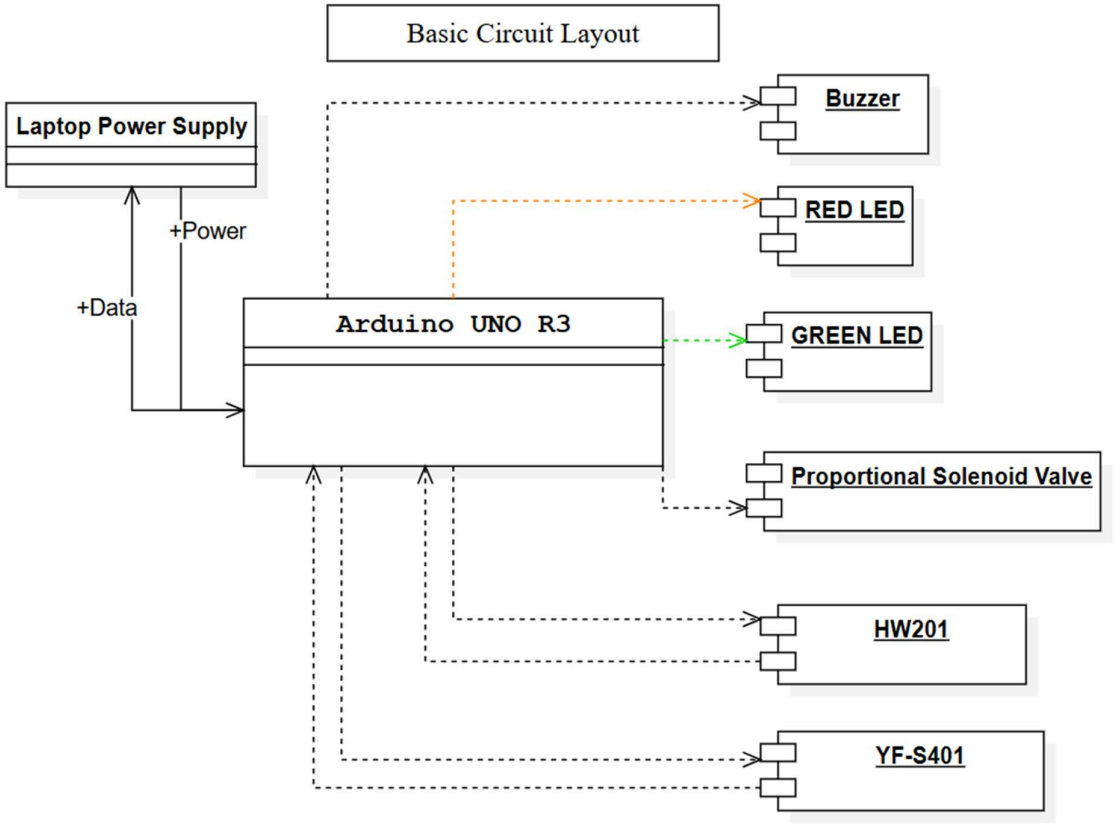
f. HW201 – IR obstacle avoidance sensor module

g. Proportional Solenoid valve

Estimated Budget

Components	Estimated Cost(in INR)
• Arduino UNO R3	• 1400-1700
• YF-S401	• 200
• HW-201	• 40
• Proportional solenoid valve	• 3500-4000
• Wirings+Resistors	• 350-500
• LEDs	• 30-40
• Buzzer	• 40

Hardware Connections



The figure elicits the proposed hardware assembly scheme, for phase 2 of the execution. Once assembled, the hardware shall be tested for:

1. Firmness of connections, and electrical response of the components.
2. Power efficiency as defined during the phase-1.